DAY03-NETWORK-INTERFACES

THEORY (10 marks)

1. What is a Network Interface Card?
   The Network interface card lets device connect and communicate on network
   It is a hardware component
   Eg: Ethernet, Wi-fi or Fiber optic cables

2. What is Ethernet?
   Wired Network technology to connect computers, routers or other devices within a local area network LAN.

3. What is RJ-45?
   It is a port/ interface with 8 position modular connector used by ethernet cables mostly found on routers, switches and PCS.

4. What does UTP stand for?
   UTP – Unshielded Twisted Pair

5. Name three common Ethernet cable categories.
   - Straight- through
   - Cross-over
   - Auto MDI-X

6. What is a straight-through cable?
   Traditionally used to connect different device types like from PC to Switch, Router to Switch.

7. What is a crossover cable?
   Used to connect similar device types like PC to PC, Switch to Switch.

8. Explain Auto MDI-X.
   Automatically detects the transmission and receiving wiring used in modern ethernet standards.

9. Difference between copper and Fiber-optic networking?

Copper	Fibre-Optic
- Lower Cost than Fiber-Optic	- Higher cost than UTP
- Maximum distance is only up to 100m	- More distance usually than UTP
- Vulnerable to EMI	- No vulnerability to EMI
- RJ - 45 Ports are used by UTP much cheaper than SFP ports	- SFP ports are more expensive than RJ-45 ports

- Leak a faint signal can lead to copy data to be known as a security risk	- No security risk

10. What is multimode Fiber?
    Multimode Fiber lets multiple light rays travel through the cable at the same time
    Shorter distance Fiber used usually in Building, data centres and campuses

11. What is single-mode Fiber?
    A type of Fiber optic cable with a very nano core that allows only single path of light to travel throughout it at a time
    Long distance Fiber used in Telecom, ISP or Network Backbones

12. Two advantages of Fiber-optic networking.
    - Transmit data as light
    - High Bandwidth
    - Long distance

13. Why is understanding network interfaces important for cybersecurity?

    Cause this is the foundation for Wireshark, Nmap, Network security, Pentesting and Incident Response

PRACTICAL IDENTIFICATION (4 marks)

13. Identify your active network adapter.
    Network Adapter : Wi-Fi
14. Write your IPv4 address, default gateway, and network adapter name.
    IPv4: 10.101.138.15
    Default Gateway: 10.101.128.1
    Network Adapter Name: Wi-Fi
    Description: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter

SCREENSHOT (3 marks)

Attach a screenshot of: ipconfig /all

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C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Kirthikaa
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-05
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 621412391
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 26-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
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DAY03-NETWORK-INTERFACES

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Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.150.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 07 October 2026 9.40.25 AM
   Lease Expires . . . . . . . . . . : 07 October 2026 11.40.24 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.150.254
   DHCPv6 IAID . . . . . . . . . . . : 201347158
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.35.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 07 October 2026 9.40.25 AM
   Lease Expires . . . . . . . . . . : 07 October 2026 11.40.24 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.35.254
   DHCPv6 IAID . . . . . . . . . . . : 234901590
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   Primary WINS Server . . . . . . . : 192.168.35.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.101.138.15(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Lease Obtained. . . . . . . . . . : 07 October 2026 8.47.34 AM
   Lease Expires . . . . . . . . . . : 07 October 2026 1.40.25 PM
   Default Gateway . . . . . . . . . : 10.101.128.1
   DHCP Server . . . . . . . . . . . : 10.101.128.1
   DHCPv6 IAID . . . . . . . . . . . : 153399993
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : 9.9.9.9
                                       208.67.222.222
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-76
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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